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School of Computer Science and Artificial Intelligence

Lab Assignment # 15.2

Program	**: B. Tech (CSE)**
Specialization	**: -**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**II**
Academic Session	**: 2025-2026**
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Enrollment No.	**: 2303A51T02**
Batch No	**: 52**
Date	**: 17/03/26**

Task 1: Setup Fast-API Backend (VS Code)

Install:

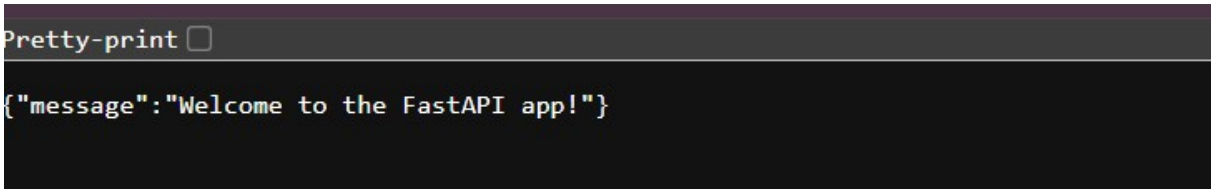
Prompt: Create a basic FastAPI app with one GET endpoint "/" that returns a welcome message.

Code:

```
from fastapi import FastAPI
app = FastAPI()
@app.get("/")
async def read_root():
    return {"message": "Welcome to the FastAPI app!"}
```

Run: python -m uvicorn main:app --reload

Output:http://127.0.0.1:8000/



Explanation: This code sets up a basic FastAPI application with endpoints to create, read, update, and delete items in a list, along with docstrings for Swagger documentation.

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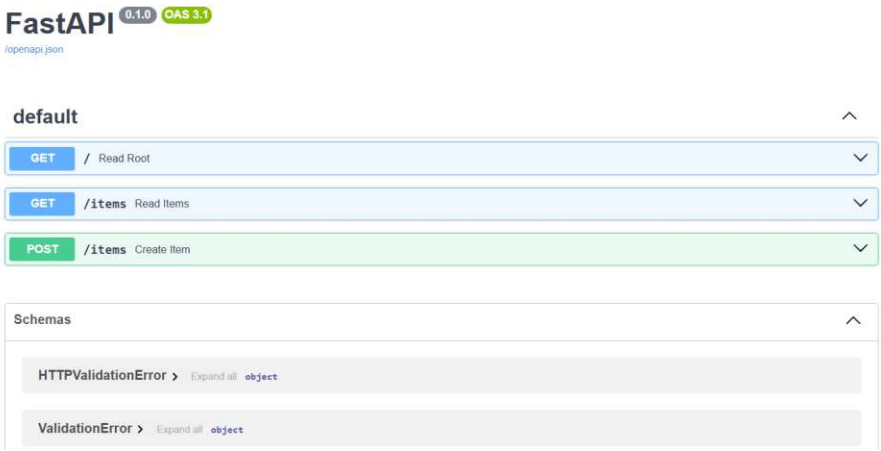
Task 2: API – Create and Read Items

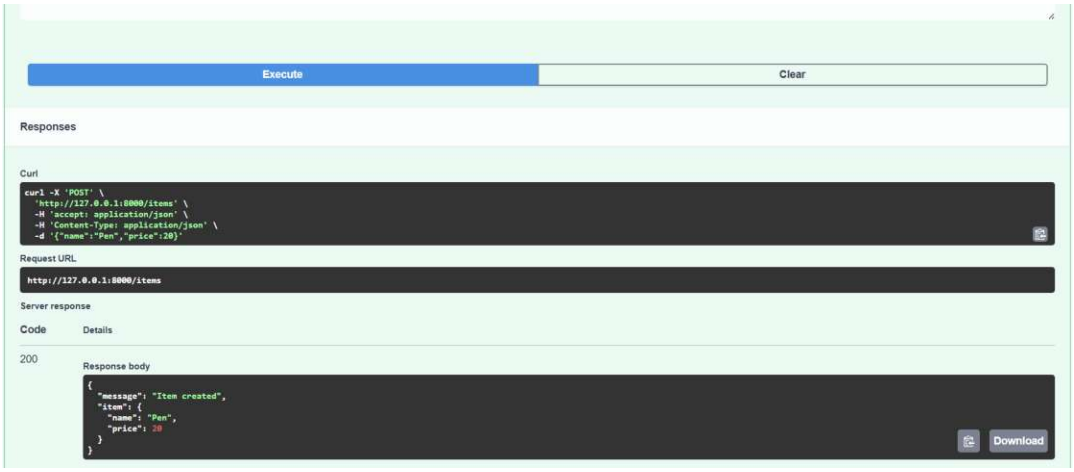
Prompt: Add GET "/items" and POST "/items" endpoints in FastAPI using a list to store items.

Code:

```python
items = []
@app.get("/items")
async def read_items():
    return items
@app.post("/items")
async def create_item(item: dict):
    items.append(item)
    return {"message": "Item created", "item": item}
```

Output:





Explanation: This code defines two endpoints in a FastAPI application to manage a list of items, allowing users to retrieve all items and add new items to the list.

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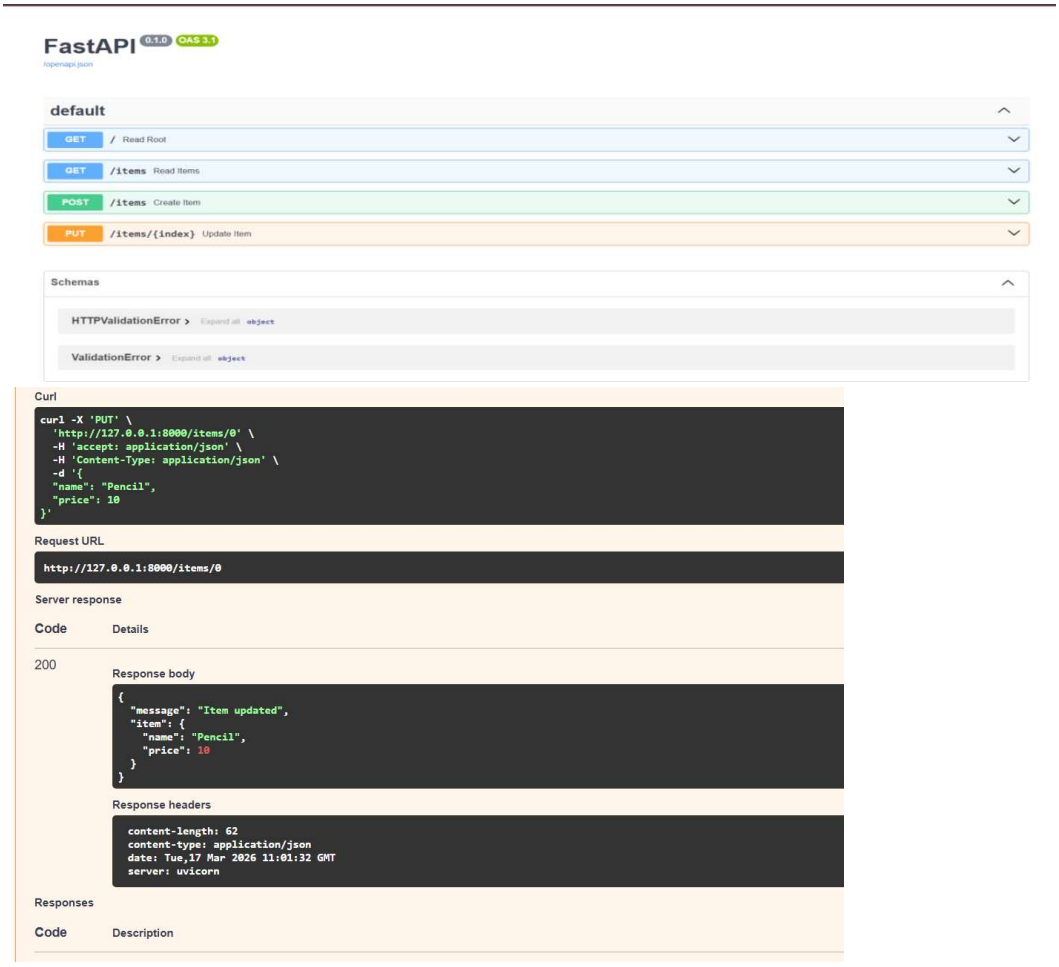
Task 3: Update Existing Item

Prompt: Add a PUT "/items/{index}" endpoint to update an item in a list by index.

Code:

```python
@app.put("/items/{index}")
async def update_item(index: int, item: dict):
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    items[index] = item
    return {"message": "Item updated", "item": item}
```

Output:



Explanation: This code adds a PUT endpoint to the FastAPI application that allows users to update an existing item in the list by specifying its index, with error handling for out-of-range indices.

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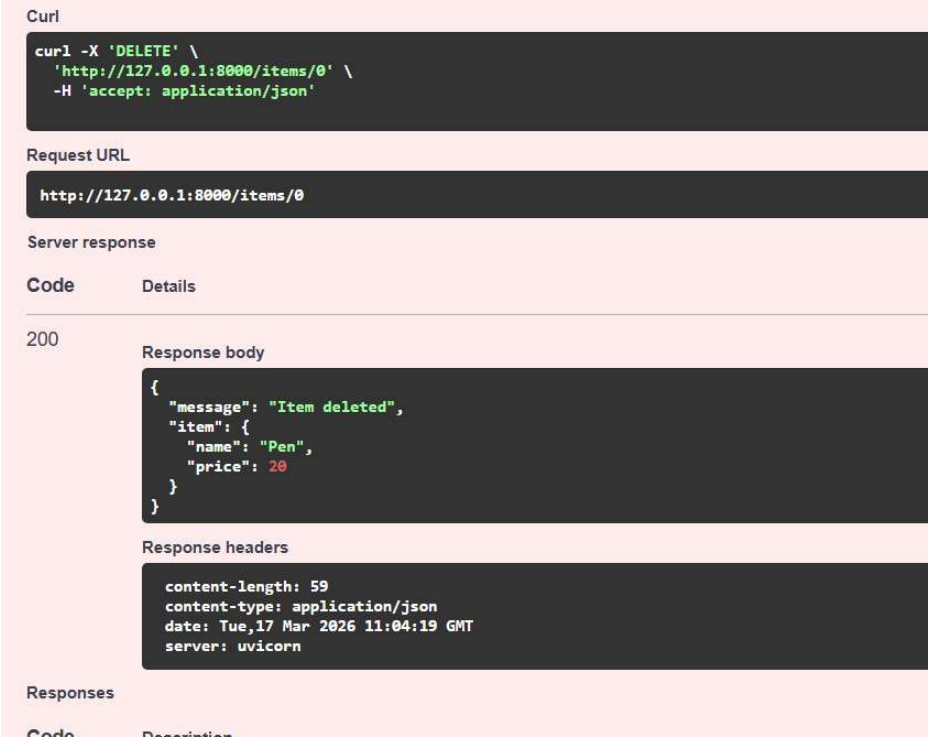
Task 4: Delete Item

Prompt: Add a DELETE "/items/{index}" endpoint to remove an item from a list.

Code:

```python
@app.delete("/items/{index}")
async def delete_item(index: int):
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    removed_item = items.pop(index)
    return {"message": "Item deleted", "item": removed_item}
```

Output:



Explanation: This code adds a DELETE endpoint to the FastAPI application that allows users to remove an item from the list by specifying its index, with error handling for out-of-range indices.

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Task 5: Auto-Generated Documentation

Prompt: Add docstrings to all FastAPI endpoints so they appear in Swagger documentation.

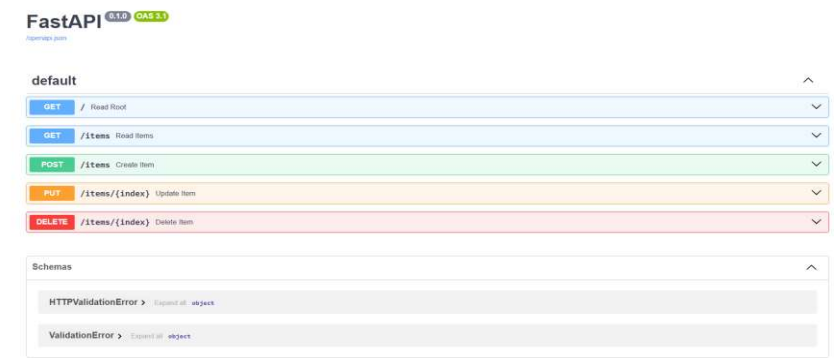
Code:

```python
@app.get("/items")
async def read_items():
    """
    Retrieve all items from the list.
    """
    return items
@app.post("/items")
async def create_item(item: dict):
    """
    Create a new item and add it to the list.
    """
    items.append(item)
    return {"message": "Item created", "item": item}
@app.put("/items/{index}")
async def update_item(index: int, item: dict):
    """
    Update an existing item in the list by its index.
    """
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    items[index] = item
    return {"message": "Item updated", "item": item}
@app.delete("/items/{index}")
async def delete_item(index: int):
    """
    Delete an item from the list by its index.
    """
    if index < 0 or index >= len(items):
        return {"error": "Index out of range"}
    removed_item = items.pop(index)
    return {"message": "Item deleted", "item": removed_item}
```

Open:

- **http://127.0.0.1:8000/docs (Swagger UI)**

Output:



Explanation: This code adds docstrings to each endpoint in the FastAPI application, providing descriptions that will be displayed in the Swagger UI for better documentation and understanding of each endpoint's functionality.